GGG-T-00563b(GSA-FSS)
January 4, 1967
INTERIM REVISION OF
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June 9, 1958

INTERIM FEDERAL SPECIFICATION

TOOLS, BLACKSMITHS' (CHISELS, FLATTERS, FULLERS, HARDIES, BOLT HEADING TOOLS, PUNCHES, AND SWAGES)

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification GGG-T-563a, dated June 9, 1958.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tools used by blacksmiths for cutting, necking, and breaking steel or metals of the same general nature; smoothing and finishing flat forgings; grooving and drawing down forgings; setting down metal to form a square corner; cutting and trimming forgings; holding bolts while forming heads; punching, clearing and enlarging holes in hot forgings; and for shaping, sizing, and smoothing round forgings.

1.1.1 Federal specification coverage. Federal specifications do not cover all types and sizes of the commodities indicated by the titles of the specifications, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The tools shall be of the following types and classes as specified (see 6.1):

Type I. - Chisel, blacksmiths'.
  Class 1. - Cold.
  Class 2. - Hot.
Type II. - Flatter, blacksmiths'.
Type III. - Fuller.
  Class 1. - Bottom, blacksmiths'.
  Class 2. - Top, blacksmiths'.
Type V. - Hardy.
Type VI. - Heading-tool, bolt.
Type VII. - Punch, blacksmiths'.
  Class 1. - Round.
  Class 2. - Square.
  Class 3. - Backing out.
Type VIII. - Swage.
  Class 1. - Bottom, blacksmiths'.
  Class 2. - Top, blacksmiths'.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
  NN-H-93 - Handles; Hickory, Striking-Tool.
  GGG-A-576 - Anvils, Blacksmiths'.

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Federal Standards:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:
MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

Department of Commerce Publication:
Simplified Practice Recommendation R17-47.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of blacksmiths' tools which are otherwise in accordance with the requirements of this specification.

3.2 Material. The tools, except for type I, class 1, and type VII, classes 1, 2, and 3 shall be forged from a good quality alloy steel or carbon steel of not less than 0.50 carbon. Type I, class 1, and type VII, classes 1, 2, and 3 shall be forged from a good quality alloy steel. The forgings shall be of uniform quality, free from burrs and flash marks.

3.3 Eyes. The handled tools shall be provided with an eye which shall extend through the head in which the handle shall be fitted. The eye shall have an inward taper at an angle of approximately 3° to permit proper wedging. The eyes shall be oval in shape, Number 2 or 3, in substantial agreement with figure 1, and in accordance with Simplified Practice Recommendation R17-47. The eye shall not buckle, crack, break, or spread under ordinary use. The median line of the eye shall be in proper alignment with the cutting, striking or punching end of the tool.

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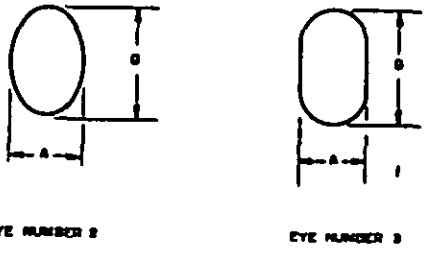


Figure 1. - Eyes.

3.4 Handles. Handled tools shall be furnished with properly fitted grade A, 16-inch (plus or minus 1 inch) hickory in accordance with HH-H-93.

3.5 Striking faces. Tool having stricking faces shall have the heads chamfered and slightly rounded or beveled to prevent chipping or spawling.

3.6 Finish. The tools shall be free from burrs, roughness, and excessive oxide scale. The tools shall have a natural finish, be oil coated; painted, or enameled.

3.7 Marking. Each tool shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.8 Type I, chisel, blacksmiths'. Chisels shall be provided with a handle in accordance ance with 3.4

3.8.1 Cutting edge. The cutting edge shall be beveled, ground to an included angle of not less than 45 nor more than 60 degrees.

3.8.2 Hardness. When tested as specified in 4.4.1, the chisel shall show a hardness value of Rockwell C 52 to C 58 on the cutting end, and Rockwell C 45 to C 55 on the striking end.

3.8.3 Class 1, cold. Class 1 chisels shall be suitable for cutting or necking cold iron and steel, and metals of the same general nature. The chisels shall be provided with either a straight cutting edge or a curved cutting edge of approximately 3-1/4 inches radius. The chisels shall conform to the dimensions shown in table I for the size specified (see 6.1), and shall be similar to figure 2.

Table I. - Type I, class 1, chisel, cold.

Overall length ± 1 - 0 D	Size of cutting edge ± 3/16 - 0 C	Size of stock at eye E		Eye Figure 1		Distance from cutting edge to center of eye ± 1/4 F	Weight without handle	
				A	B			
		Min.	Max.	Min.	Min.		Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inches	Inches	Pounds	Pounds
5-1/2	1-5/32	1-3/16	1-5/16	5/8	7/8	3-7/16	1-3/4	2-1/4
6-1/2	1-13/32	1-7/16	1-9/16	3/4	1	4	2-3/4	3-1/4
7-1/2	1-21/32	1-11/16	1-13/16	1	1-1/4	4-5/8	4-1/2	5-1/2

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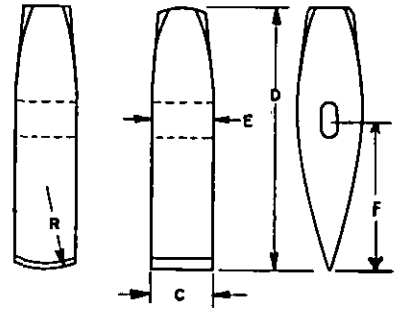


Figure 2. - Type I, class 1 chisel, blacksmiths', cold.

3.8.4 Class 2, hot. Class 2 chisels shall be suitable for cutting hot iron and steel and other metals of the same general nature. The chisels shall conform to the dimensions shown in table II for the size specified (see 6.1) and shall be similar to figure 3.

Table II. - Type I, class 2, chisel, blacksmiths', hot.

Overall length ± 1 - 0 D	Size of cutting edge ± 3/16 - 0 C	Size of stock at eye E		Eye Figure 1		Distance from cutting edge to center of eye ± 1/4 F	Weight without handle	
				A	B			
		Min.	Max.	Min.	Min.		Min.	Max.
Inches	Inches	Inches	Inches	Inch	Inches	Inches	Pounds	Pounds
6-7/8	1-15/32	1-3/16	1-5/16	5/8	7/8	4-13/16	1-3/4	2-1/4
7-3/4	1-27/32	1-7/16	1-9/16	3/4	1	5-1/8	2-3/4	3-1/2
9	2-3/32	1-11/16	1-13/16	1	1-1/4	6	4-1/2	5-1/2

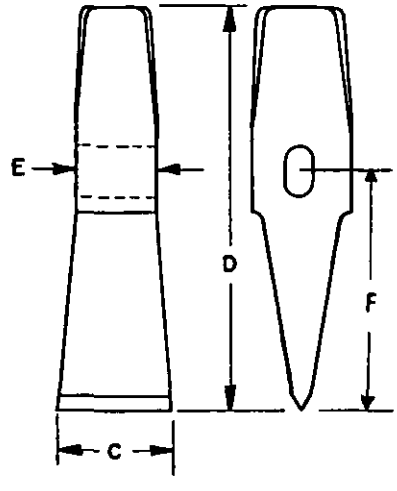


Figure 3. - Type I, class 2 chisel, blacksmiths', hot.

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3.9 Type II flatter, blacksmiths'. Flatters shall be suitable for smoothing and finishing flat forgings. Each flatter shall be provided with a handle in accordance with 3.4. The face of the flatter shall be square, ground flat, and be uniform in thickness. The flatter shall conform to the dimensions shown in table III for the size specified (see 6.1), and shall be similar to figure 4.

Table III. - Type II flatter, blacksmiths'.

Overall length ℄ 1 - 0 D	Size of face ℄ 3/16 - 0 C	Eye Figure 1 A Min.	Eye Figure 1 B Min.	Distance from striking head to center of eye ℄ 1/4 E	Weight without handle Min.	Weight without handle Max.
Inches	Inches	Inch	Inches	Inches	Pounds	Pounds
4	1-5/32	5/8	7/8	2-9/16	1-1/2	2
4-1/2	1-13/32	5/8	7/8	2-15/16	2	3-1/4
5	1-21/32	3/4	1	3-1/16	2-3/4	4-1/2
4	1-7/8	5/8	7/8	2-1/16	2	3
5	2-3/8	3/4	1	2-1/8	2-3/4	4-1/4
5-1/2	2-7/8	3/4	1-3/8	2-7/8	4-3/4	7
5-1/2	3-7/8	3/4	1-3/8	2-13/16	6	8-1/4

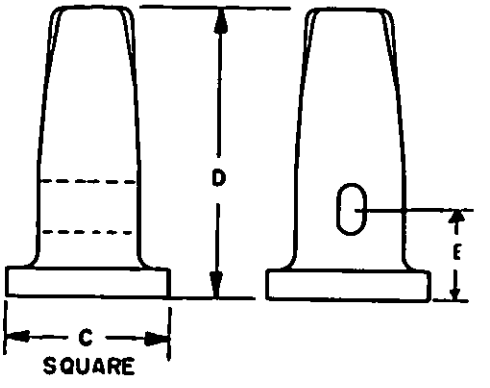


Figure 4. - Type II flatter, blacksmiths'.

3.9.1 Hardness. When tested as specified in 4.4.1, the flatter shall show a hardness value of Rockwell C 45 to 55 on the striking end.

3.9.1 Hardness. When tested as specified in 4.4.1, the flatter shall show a hardness value of Rockwell C 44 to 55.

3.10 Type III fuller. Fullers shall be suitable for necking, grooving, and drawing down forgings.

3.10.1 Hardness. When tested as specified in 4.4.1, the fuller shall show a hardness value of Rockwell C 44 to 55 on the convexed end of the tool.

3.10.2 Class 1 bottom, blacksmiths'. Class 1 fuller shall be provided with a square shank, shoulder, and a convex radius. Unless otherwise specified, the shank shall be one inch square plus 0, minus 1/8 inch, and length shall be 2-3/8 inches plus or minus 1/2 inch. The face of the shoulder shall be flat and square with the shank. Shoulder shall be of sufficient width to provide a stable working surface when shank is placed in hardie hole of anvil as covered by GGG-A-576. The convex radius shall be accurately and smoothly finished. The fuller shall conform to the dimensions shown in table IV for the size specified (see 6.1), and shall be similar to figure 5.

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Table IV. - Type III, class 1 fuller, bottom, blacksmiths'.

Nominal Size	Width W		Radius R		Length D		Weight	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inch	Inch	Inches	Inches	Pounds	Pounds
1/4	2-1/4	2-1/2	7/64	9/64	4	5	1-1/2	2-1/4
3/8	2-1/4	2-1/2	11/64	13/64	4	5	1-1/2	2-1/4
1/2	2-1/4	2-1/2	15/64	17/64	4	5-1/2	1-1/2	2-1/2
3/4	2-1/4	2-1/2	23/64	25/64	4	5-1/2	2	2-3/4
1	2-1/4	2-1/2	15/32	17/32	4	5-1/2	2-1/4	3
1-1/4	- -	- -	19/32	21/32	4-1/8	5-1/2	3	3-1/4
1-1/2	- -	- -	23/32	25/32	4-1/8	5-1/2	3	3-1/2
1-3/4	- -	- -	27/32	29/32	4-1/4	5-1/2	3-1/2	4
2	- -	- -	31/32	1-1/32	4-1/4	5-1/2	3-1/2	4-1/2

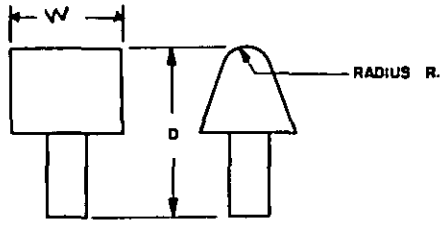


Figure 5. - Type III, class 1 fuller, bottom, blacksmiths'.

3.10.3 Class 2, fuller, top, blacksmiths'. Class 2 fuller shall be provided with a handle in accordance with 3.4. The convex radius shall be accurately and smoothly finished. The fuller shall conform to the dimensions shown in table V for the size specified (see 6.1) and shall be similar to figure 6.

Table V. - Type III, class 2, fuller, top, blacksmiths'.

Nominal Size	Width W		Radius R		Length D		Eye Figure 1		Distance from striking head to center of eye ± 1/4 C	Weight Without Handle	
	Min.	Max.	Min.	Max.	Min.	Max.	A Min.	B Max.		Min.	Max.
Inches	Inches	Inches	Inch	Inch	Inches	Inches	Inch	Inch	Inches	Pounds	Pounds
1/4	2-1/4	2-1/2	7/64	9/64	4-1/4	5-1/2	1/2	7/8	2-7/16	2	2-1/4
3/8	2-1/4	2-1/2	11/64	13/64	4-1/4	5-1/2	5/8	7/8	2-7/16	2	2-3/4
1/2	2-1/4	2-1/2	15/64	17/64	4-1/4	5-1/2	5/8	7/8	2-7/16	2	2-3/4
3/4	2-1/4	2-1/2	23/64	25/64	4-3/4	6-1/2	5/8	7/8	2-1/2	2-1/2	2-3/4
1	2-1/4	2-1/2	15/32	17/32	5-1/4	6-3/4	5/8	1	2-3/8	2-3/4	3
1-1/4	- -	- -	19/32	21/32	5-1/4	7	3/4	1	- -	3-1/4	4
1-1/2	- -	- -	23/32	25/32	5-1/4	7	3/4	1	- -	4	4-1/2
1-3/4	- -	- -	27/32	29/32	5-1/4	7	3/4	1	- -	4-1/4	4-1/2
2	- -	- -	31/32	1-1/32	5-1/4	7	3/4	1	- -	4-1/4	4-1/2

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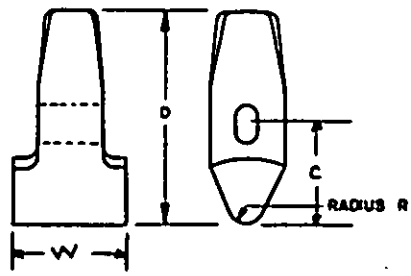


Figure 6. - Type III, class 2 fuller, top, blacksmiths'.

3.11 Type V, hardy. The hardy shall be suitable for cutting and trimming forgings and shall be provided with a square shank and a shoulder. The shank shall be 2-3/8 inches plus or minus 1/2 inch in length. The face of the shoulder shall be flat and square with the shank. Shoulder shall be of sufficient width to provide a stable working surface when the shank is placed in hardie hole of anvil as covered by GGG-A-576. The tool shall have a gradual taper from the cutting edge to the shoulder. The cutting edge shall be ground sharp and flat.

3.11.1 The hardy shall conform to the dimensions shown in table VI for the size specified (see 6.1) and shall be similar to figure 7.

Table VI.- Type V, hardy.

Size of shank / .000 - .0625 A	Overall length (min.) C	Width of bit B		Weight	
		Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Pounds	Pounds
1/2	5	1-17/32	1-23/32	5/8	1
1/2	5	1-5/8	1-3/4	7/8	1-1/4
3/4	5	1-25/32	1-31/32	1-1/8	1-1/2
7/8	5	1-29/32	2-3/32	1-1/2	2-1/8
7/8	5	2-1/32	2-7/32	1-3/4	2-3/8
1-1/4	6	2-1/8	2-11/32	2-1/8	3
1-1/4	6	2-1/8	2-1/2	2-1/4	3-1/4

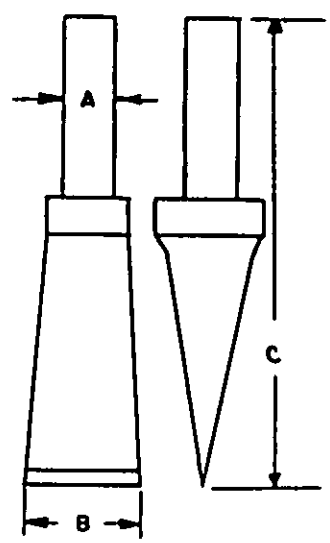


Figure 7. - Type V, hardy.

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3.11.2 Hardness. The cutting edge of the hardy shall show not less than 49 nor more than 54 on a Rockwell C scale not less than 3/8 inch back from the cutting edge when tested as specified in 4.4.1.

3.12 Type VI, heading tool, bolt. The heading-tool shall be suitable for holding bolts while forming the heads. The head of the tool shall be round on the sides, the faces flat and uniform in thickness. The hole in the tool shall be true, round, and concentric with the head. The tool shall conform to the dimensions shown in table VII for the diameter specified (see 6.1) and shall be similar to figure 8.

Table VII. - Type VI, heading-tool, bolt.

Size of hole ± 3/64 A	Length B		Diameter C		Thickness D		Weight	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inch	Inches	Inches	Inches	Inches	Inches	Inches	Pounds	Pounds
1/4	13	16-1/2	2	2-1/2	1-1/8	1-3/8	2-1/2	3-1/2
3/8	13	16-1/2	2	2-1/2	1-1/8	1-3/8	2-1/2	3-1/2
1/2	13	16-1/2	2	2-1/2	1-1/8	1-3/8	2-1/2	3-1/2
5/8	13-1/2	16-1/2	2	2-1/2	1-1/8	1-3/8	2-1/2	3-1/2
3/4	14-1/2	20	2-1/4	2-3/4	1-1/4	1-1/2	3-1/4	4-3/4
7/8	14-1/2	20	2-1/4	2-3/4	1-1/4	1-1/2	3-1/4	4-3/4
1	14-1/2	20	2-1/4	2-3/4	1-1/4	1-1/2	3-1/4	4-3/4

3.12.1 Hardness. When tested as specified in 4.4.1, the bolt heading tool shall show a Rockwell C hardness of 44 to 55 on the head of the tool.

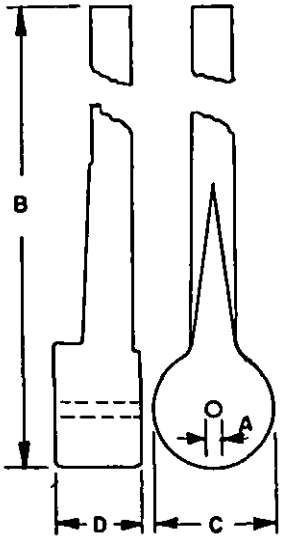


Figure 8. - Type VI, heading-tool, bolt.

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3.13 Type VII, punch, blacksmiths'. Punches, except for class 3, shall be suitable for punching, clearing, and enlarging holes in hot forgings. Class 3 punch shall be suitable for backing other punches and removing metal plugs. Each punch shall be provided with a handle in accordance with 3.4. Punching end shall be ground concentric with the axis of the body showing a runout of not more than 0.020 inch total indicator reading.

3.13.1 Hardness. When tested as specified in 4.4.1, the alloy steel punch shall show a hardness value of Rockwell C 50 to 55 on the punch end and 40 to 45 on the striking end.

3.13.2 Class 1, round. The punching end of class 1 punch shall be circular in shape and tapered from the shoulder to the punch end. Punch shall conform to the dimensions shown in table VIII for the size specified (see 6.1), and shall be similar to figure 9.

Table VIII. - Type VII, class 1 punch, blacksmiths', round.

Size ℄ 3/16 C	Overall length ℄ 1 - 0 D	Length of punch E		Size of stock at eye F		Distance from striking end to center of eye ℄ 1/4 O	Eye A	Eye B	Weight without handle	
		Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch	Pounds	Pounds
1/4	6-1/2	3	3-3/4	7/8	1-3/8	2-1/4	5/8	7/8	1	1-1/2
3/8	6-3/4	3-1/4	4	7/8	1-3/8	2-3/8	5/8	7/8	1-1/4	1-3/4
1/2	7	3-1/4	4	1	1-1/2	2-3/4	5/8	7/8	1-3/4	2-1/4
5/8	7-1/4	3-1/4	4	1-1/8	1-5/8	2-7/8	5/8	1	2	2-1/2
3/4	7-1/2	3-1/2	4-1/4	1-1/8	1-5/8	2-7/8	5/8	1	2-1/4	2-3/4
7/8	7-3/4	3-1/2	4-1/4	1-1/4	1-3/4	3-3/16	5/8	1	2-3/4	3-1/2
1	8	3-1/2	4-1/4	1-1/4	1-3/4	3-3/16	5/8	1	3	4-1/2

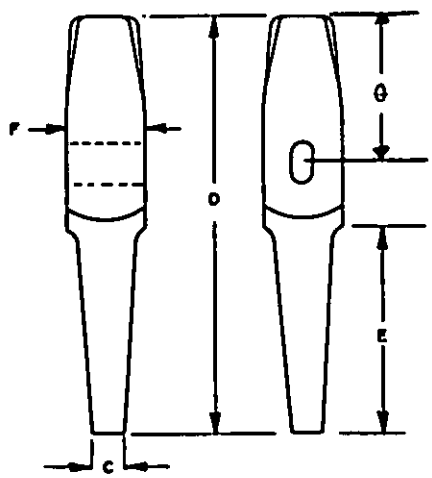


Figure 9. - Type VII, class 1 punch, blacksmiths', round.

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3.13.3 Class 2, square. The punching end of class 2 punch shall be square in shape and tapered from the shoulder to the punch end. Punch shall conform to the dimensions shown in table IX for the size specified (see 6.1), and shall be similar to figure 10.

Table IX. - Type VII, class 2 punch, blacksmiths', square.

Size ± 3/64 C	Overall length ± 1 - 0 D	Length of punch E		Size of stock at eye F		Distance from striking end to center of eye ± 1/4 G	Eye Figure 1 A	Eye Figure 1 B	Weight without handle	
		Min.	Max.	Min.	Max.		Min.	Min.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch	Pounds	Pounds
1/4	6-1/2	2-3/4	4-1/4	7/8	1-3/8	2-1/4	5/8	7/8	1	1-3/4
3/8	6-1/2	3	4-1/4	7/8	1-3/8	2-3/8	5/8	7/8	1-1/4	2-1/2
1/2	7-3/4	3	4-1/4	1	1-1/2	2-3/4	3/4	1	2	2-1/2
5/8	6-3/4	3	4-1/4	1-1/8	1-5/8	3-1/4	3/4	1	2-1/4	2-1/2
3/4	7	3	4-1/4	1-1/8	1-5/8	3-1/4	3/4	1	2-1/2	3
7/8	7	2-7/8	4-1/4	1-1/4	1-3/4	3-3/8	3/4	1	3	3-1/2
1	8	2-7/8	4-1/4	1-1/4	1-3/4	3-3/8	3/4	1	3	4

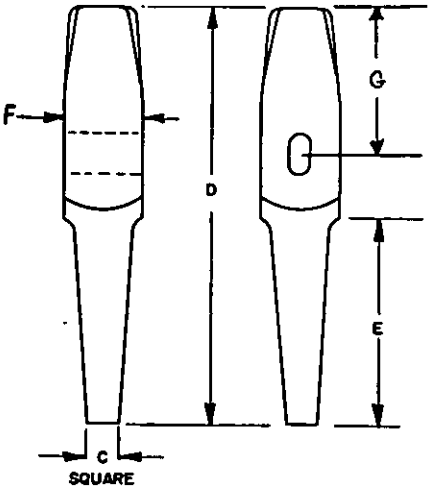


Figure 10. - Type VII, class 2, punch, blacksmiths', square.

3.13.4 Class 3, backing out. The punch end of class 3 punch shall be circular in shape and straight from the shoulder to the punch end. Punch shall conform to the dimensions shown in table X for the size specified (see 6.1) and shall be similar to figure 11.

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Table X. - Type VII, class 3 punch, backing out.

Size ± 3/64 C	Overall length ± 1 - 0 D	Length of punch E		Size of stock at eye F		Distance from striking end to center of eye ± 1/4 - 0	Eye Figure 1 A		Weight without handle	
		Min.	Max.	Min.	Max.		Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch	Pounds	Pounds
3/8	6-3/4	3	4	7/8	1-3/8	2-7/8	5/8	7/8	1-1/8	1-3/4
1/2	7	3	4	1	1-1/2	2-5/8	3/4	1	1-3/4	2
5/8	7-1/4	3	4	1-1/8	1-5/8	2-5/8	3/4	1	2	2-1/4
3/4	7-1/2	3-1/8	4-5/8	1-1/8	1-5/8	2-7/8	3/4	1	2-1/4	3
7/8	7-3/4	3-1/4	4-9/16	1-1/4	1-3/4	2-7/8	3/4	1	2-3/4	3-1/2
1	8	3-5/8	4-7/8	1-1/4	1-3/4	2-7/8	3/4	1	3	4

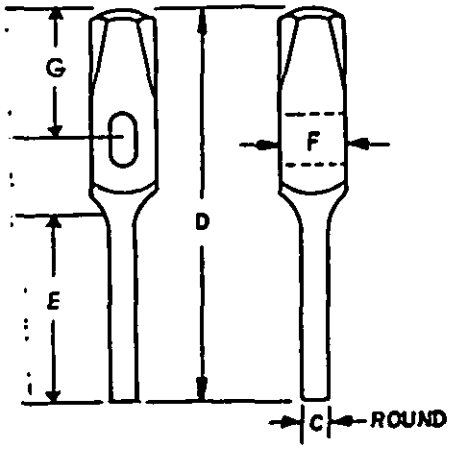


Figure 11. - Type VII, class 3 punch, blacksmiths', backing out.

3.14 Type VIII, swage. Swages shall be suitable for shaping, sizing, and smoothing round forgings. The concave radius shall be accurately and smoothly finished. The depth of the grooves shall be slightly less than the radius "R" dimension to prevent pinching of the worked metal.

3.14.1 Hardness. When tested as specified in 4.4.1, the swage shall show a hardness value of Rockwell C 35 to 45 on the head of the tool.

3.14.2 Class 1, bottom, blacksmiths'. Class 1 swage shall be provided with a square shank, and a shoulder. The shank shall be 2-3/8 inches, plus or minus 1/2 inch, in length and, unless otherwise specified, 1 inch square, plus 0, minus 1/8 inch. The face of the shoulder shall be flat and square with the shank. The shoulder shall be of sufficient width to provide a stable working surface when shank is placed in hardie hole of anvil as covered by GGG-A-576. The tool shall conform to the dimensions shown in table XI for the size specified (see 6.1) and shall be similar to figure 12.

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Table XI. - Type VIII, class 1 swage, bottom, blacksmiths'.

Size ≠ 3/64	Width W		Radius R		Length D		Weight	
	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Pounds	Pounds
1/4	1-7/8	2-1/8	7/64	5/32	3-1/4	4	2	2-1/4
3/8	1-7/8	2-1/8	11/64	7/32	3-1/4	4-3/4	2	2-1/4
1/2	1-7/8	2-1/8	15/64	9/32	3-1/4	4-3/4	2	2-1/2
5/8	2-1/8	2-3/8	19/64	11/32	3-1/2	4-1/2	2-1/4	2-1/2
3/4	2-1/8	2-3/8	23/64	13/32	3-1/2	4-1/2	2-1/4	2-1/2
7/8	2-1/4	2-1/2	27/64	15/32	3-1/2	4-1/2	2-1/2	2-3/4
1	2-1/4	2-1/2	15/32	35/64	3-1/2	4-1/2	2-1/2	3
1-1/4	- -	- -	19/32	43/64	3-1/2	5	3	3-1/2
1-1/2	- -	- -	23/32	51/64	3-5/8	5	3	3-1/2
1-3/4	- -	- -	27/32	59/64	3-3/4	5-1/4	3-3/4	4-1/2
2	- -	- -	31/32	1- 3/64	3-3/4	5-1/2	3-3/4	4-1/2
2-1/2	- -	- -	1- 7/32	1-17/64	4	5-1/2	4-1/2	5-1/2
3	- -	- -	1-15/32	1-35/64	4-1/4	6	4-1/2	5-1/2

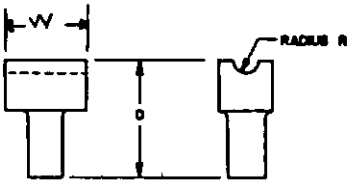


Figure 12. - Type VIII, class 1 swage, bottom, blacksmiths'.

3.14.3 Class 2, top, blacksmiths'. Class 2 swage shall be provided with a handle in accordance with 3.4. The tool shall conform to the dimensions shown in table XII for the size specified (see 6.1), and shall be similar to figure 13.

Table XII. - Type VIII, class 2, swage, top, blacksmiths'.

Nominal size	Radius R		Length D		Width W		Eye Figure 1		Weight without handle	
	Min.	Max.	Min.	Max.	Min.	Max.	A	B		
							Min.	Min.	Min.	Max.
Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inch	Inch	Pounds	Pounds
1/4	7/64	5/32	4	5	2-1/4	2-1/2	1/2	7/8	1	2
3/8	11/64	7/32	4	5-1/4	2-1/4	2-1/2	1/2	7/8	1-1/2	2-1/2
1/2	15/64	9/32	4	5-3/4	2-1/4	2-1/2	1/2	7/8	1-1/2	2-1/2
5/8	19/64	11/32	4-1/4	5-1/4	2-1/2	2-3/4	1/2	7/8	2	3
3/4	23/64	13/32	4-1/4	5-1/2	2-1/2	2-3/4	5/8	7/8	2	3-1/4
7/8	27/64	15/32	4-1/4	5-1/2	2-5/8	2-7/8	5/8	7/8	2-1/2	3-1/4
1	15/32	35/64	4-1/4	5-1/2	2-5/8	2-7/8	5/8	1	2-1/2	3-1/4
1-1/4	19/32	43/64	4-1/4	5-3/4	- -	- -	3/4	1	3	3
1-1/2	23/32	51/64	4-1/2	5-7/8	- -	- -	3/4	1	3-1/4	4-1/4
1-3/4	27/32	59/64	5	6	- -	- -	3/4	1	4	4-3/4
2	31/32	1 3/64	5	6	- -	- -	3/4	1	4	4-3/4
2-1/2	1 7/32	1 19/64	5	6	- -	- -	3/4	1	4-1/2	5-1/2
3	1 15/32	1 35/64	5	6-1/4	- -	- -	3/4	1	4-1/2	5-1/2

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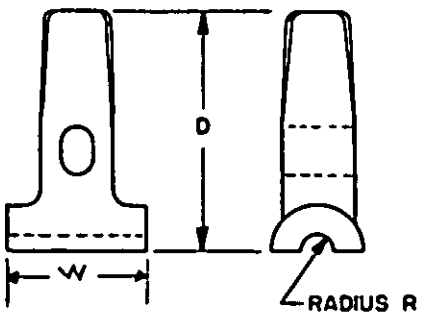


Figure 13. - Type VIII, class 2 swage, top, blacksmiths'.

3.15 Workmanship. Workmanship shall be of the highest grade throughout and equal in every respect to good commercial practice. Blacksmiths' tools shall be free from imperfections that may affect their serviceability or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table XIII.

Table XIII. - Sampling data.

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed in Terms of	Reference
Visual examination	One each	II	2.5	Defects per 100 units	4.3.1
Dimensional and weight examination	One each	S-3	2.5	Defects per 100 units	4.3.2
Testing	One each	S-2	4.0	Defects per 100 units	4.4
Preparation for delivery	One container	S-2	4.0	Defects per 100 units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship and marking.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.4.1 through 4.4.2.3.

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4.4.1 Hardness test. The hardness test shall be in accordance with the applicable requirements of method 243.1 of Fed. Test Method Std. No. 151 and the value thus obtained shall comply with the requirements of 3.8.2, 3.9.1, 3.10.1, 3.11.1, 3.13.1, and 3.14.1. Unless otherwise specified, impressions shall be taken 1/4 inch from the cutting or striking edge.

4.4.2 Performance test.

4.4.2.1 Specimens used for performance test: type I, chisel and type V, hardy. A round steel bar showing a hardness of not less than 35 nor more than 40 on a Rockwell C scale shall be used in subjecting the chisels and hardies to the performance test.

4.4.2.2 Type I, class 1, cold, and class 2, hot, chisels. The sample chisels shall be held in as near a vertical position as possible with the work end resting on the surface of the bar. The chisels shall be struck as squarely as possible a sufficient number of blows to produce an indentation 1/8 inch deep. The class 1 chisel shall make twelve indentations in the test bar. The class 2 chisel shall make six indentations in the test bar. The cutting edge and the head of the chisels shall show no chipping or undue wear.

4.4.2.3 Type V, hardy. Sample hardies shall be set in a vertical position with the shoulder against a solid foundation. The bar shall be held at right angles to the cutting edge of the hardy and struck a sufficient number of blows by a heavy hammer to produce an indentation of 1/8 inch. Six indentations shall be made without the cutting edge of the hardies chipping or showing undue wear.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table XIV. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Table XIV. - Classification of preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation and packaging.

5.1.1 Level A. Unless otherwise specified (see 6.1), level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 Level C. Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

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5.2 Packing.

5.2.1 Level A. Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 Level B. Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 Level C. Tools, preserved and packaged as specified in 5.1.1 or 5.1.2 as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination.

5.3 Marking.

5.3.1 Military agencies. In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of tools required (see 1.2).
(c) Size of tools required (see applicable table).
(d) Specify the size of square shank, if different (see 3.10.2 and 3.14.1).
(e) Selection of applicable level of preservation, packaging and packing required (see 5.1 and 5.2).

6.2 For civil agency procurements. When level B packaging is required for civil agency procurements, the requirements in 5.1.1, level A packaging shall apply.

6.3 Tools covered by this specification are in accordance with Simplified Practice Recommendation R17-47, where applicable.

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